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1. Perform recursive Function binary search using C.

```
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int key){
    if(low > high){
        return -1;
    }

    int mid = (low + high) / 2;

    if(arr[mid] == key){
        return mid;
    }
    else if(key < arr[mid]){
        return binarySearch(arr, low, mid - 1, key);
    }
    else{
        return binarySearch(arr, mid + 1, high, key);
    }
}

int main(){
    int arr[100], n, key, res;
    int i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter sorted elements: ");
    for(i = 0; i < n; i++){
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);
```

```
    res = binarySearch(arr, 0, n - 1, key);

    if(res == -1)
        printf("Element not found\n");
    else
        printf("Element found at index %d\n", res);

    return 0;
}
```